

Cybersecurity & Cyber-Resilience Strategy for the Sovereign Cloud–Edge–IoT Continuum

Iraklis Symeonidis

March 27, 2026

Abstract

The rapid decentralization of computing infrastructures toward the Cloud–Edge–IoT continuum introduces unprecedented cybersecurity challenges, including expanded attack surfaces, heterogeneous environments, and increased dependency on non-sovereign technologies. This paper proposes a structured, multi-stage methodology for defining and prioritizing cybersecurity capabilities required to achieve a secure and sovereign European computing continuum. The approach combines a pipeline-oriented transformation model with a seven-dimensional analytical framework, enabling traceable transitions from architectural capabilities to strategic priorities and policy-aligned recommendations. A four-filter prioritization logic identifies critical capabilities based on regulatory urgency, architectural dependencies, sovereignty risks, and market feasibility. These priorities are further analyzed through a progression matrix capturing operational deployment and strategic sovereignty evolution across time phases. The methodology provides a reproducible and evidence-based framework for guiding research, innovation investments, and policy decisions in the context of the European digital sovereignty agenda.

Keywords: *Cybersecurity, Cloud-Edge-IoT, Digital Sovereignty, Zero Trust, Post-Quantum Cryptography, EU Policy, Cyber-Resilience*

1 Introduction

1.1 Contributions

This paper makes the following contributions:

- A structured pipeline methodology for cybersecurity capability prioritization in the Cloud–Edge–IoT continuum.
- A 4-dimensional filtering logic linking regulatory, architectural, sovereignty, and market-adoption considerations.
- A hybrid 7-dimensional profiling framework for actionable R&I prioritization.
- A progression matrix model capturing operational deployment and strategic sovereignty evolution across time phases.
- A traceable transformation from technical capabilities to policy-aligned strategic recommendations.

This manuscript/roadmap defines the architectural guardrails required to secure the Cloud-Edge-IoT Continuum. It moves beyond traditional “perimeter security” toward a model of Cyber-Resilience—where systems are engineered to proactively anticipate, withstand, recover from, and adapt to attacks while maintaining strict Digital Sovereignty. **Approach:** To provide actionable, risk-aware Research & Innovation (R&I) priorities, the strategy is organized into a two-part framework:

Purpose of This Document

This roadmap provides a structured methodology for European policymakers, research organizations, and industry stakeholders to:

- Identify and prioritize cybersecurity R&I investments for the Cloud-Edge-IoT continuum
- Track capability maturity using objective Operational Deployment Phase (ODP) and Strategic Sovereignty Level (SSL) metrics
- Align Horizon Europe funding with strategic autonomy goals and regulatory compliance requirements
- Ensure coordinated implementation across Member States and European initiatives (ECCC, Digital Europe Programme, EU Chips Act)

Target Audiences

European Commission & Funding Bodies:

Use the 4-Filter Prioritization Framework to allocate Horizon Europe, Digital Europe Programme, and ECCC funding to capabilities with the highest strategic impact.

National Cybersecurity Agencies: Leverage the progression matrices to coordinate national R&I investments with EU-wide timelines and ensure interoperability across Member States.

Research Consortia & Industry: Utilize the 7-Dimensional Analysis template to structure

project proposals, clearly articulating baseline, target, drivers, risks, and alignment with European strategic goals.

Standardization Bodies (CEN/CENELEC, ETSI):

Reference the architectural framework and MIMs to develop harmonized standards for secure, interoperable computing continuum infrastructure.

How to Navigate This Document

1. **Start with Section 1-3:** Understand the threat landscape, regulatory drivers, and strategic context for securing the European computing continuum.
2. **Review Section 4:** Familiarize yourself with the 6 Security Layers and 4 Cross-Layers comprising the architectural framework.
3. **Apply Section 5:** Use the 4-Filter Prioritization Framework to assess which capabilities are critical for your specific context (national strategy, industrial sector, research focus).
4. **Study Section 6-7:** Understand the 7-Dimensional Analysis methodology and ODP/SSL progression metrics before designing projects or policies.
5. **Reference Section 9 (Priorities 1-3):** Use the detailed capability profiles as templates for structuring additional priorities relevant to your domain.
6. **Implement Section 10:** Actionable strategic recommendations for Phase 1 (2025-2030) provide concrete next steps.

Key Principles

- **Evidence-Based Prioritization:** All priorities must pass the 4-Filter Framework (Regulatory Urgency, Foundational Dependency, Sovereignty Cliff, Market Feasibility).

- **Sovereignty-First Design:** Investments should systematically reduce foreign dependencies (measured via SSL metrics) while building European industrial capacity.
- **Compliance-by-Design:** All R&I outputs must align with or anticipate NIS2, CRA, AI Act, and Data Act requirements to ensure market relevance.
- **Federated Interoperability:** Solutions must prioritize standardization and open architectures (e.g., Simpl middleware, open-source RISC-V) over proprietary, siloed approaches.
- **Measurable Progression:** Avoid subjective TRL assessments; instead, use objective ODP (market integration) and SSL (sovereignty) metrics with clear phase transition criteria.

2 Background & Driving Factors

Destinations:

- **Primary:** 4. A secure, sovereign European computing continuum infrastructure.
- **Secondary:** 6. European leadership in emerging disruptive computing paradigms.
- **Decentralization Risk:** The paradigm shift from the centralized Cloud to a distributed Cloud-Edge-IoT continuum radically expands the attack surface. This creates environments that are often “too many, too diverse, and too dispersed” to secure with traditional perimeter defenses.
- **Physical Exposure:** Edge nodes and far-edge IoT devices frequently operate in the field. Unlike hyperscaler data centers, they lack strict physical and environmental security controls, making them highly vulnerable to tampering.
- **Regulatory Pressure:** There is an immediate, complex need to operationalize the NIS2 Directive [1], the Cyber Resilience Act (CRA) [2], and

the AI Act [3] across a highly fragmented technological ecosystem to ensure compliance and market access.

- **Digital Sovereignty:** Europe must urgently reduce its systemic reliance on non-EU technology providers. Doing so will mitigate extra-territorial data access risks and avoid geopolitical vendor lock-in, ensuring independent control over critical infrastructure.

2.1 Europe in Relation to the State of the Art

Comparison vs. Hyperscalers:

- **EU Strengths:** Europe leads globally in Trust, Privacy-by-Design, proactive regulation (GDPR, Data Act, AI Act), and rigorous, standardized certification frameworks (e.g., EUCS [4]).
- **EU Gaps:** The European market suffers from a lack of unified, hyper-scale security orchestration. Furthermore, there is a critical dependency on non-EU hardware (secure silicon) and highly fragmented threat intelligence sharing among member states.

Strategic Goal: To definitively transition the European computing continuum from a state of “reactive, siloed security” to an advanced architecture defined by “autonomous, federated, and certified Security-by-Design.”

3 Related Work

Existing research on cybersecurity for distributed computing infrastructures has primarily focused on isolated domains such as cloud security, edge computing protection, and IoT security frameworks. However, limited work addresses the holistic security challenges of the integrated Cloud-Edge-IoT continuum.

Recent initiatives, including the European Union Cybersecurity Strategy, ENISA reports, and Horizon Europe projects, emphasize the need for sovereignty-driven security architectures. Prior studies on Zero

Trust Architectures (ZTA), Post-Quantum Cryptography (PQC), and confidential computing provide important building blocks but lack a unified prioritization and transition methodology.

This work differentiates itself by proposing a structured, end-to-end methodological pipeline that connects architectural capabilities with policy-driven prioritization and phased implementation strategies.

4 Proposed Methodology for Cybersecurity Prioritization in the Computing Continuum

5 Methodology Validation and Application

To demonstrate the applicability of the proposed methodology, selected priorities such as Hardware Roots of Trust and PQC Agility are evaluated across the defined progression matrix.

The validation focuses on:

- Consistency of prioritization outcomes using the 4-filter logic
- Traceability from capability definition to strategic recommendation
- Alignment with EU regulatory frameworks, including NIS2, CRA, and the AI Act

The results confirm that the methodology enables:

- Systematic identification of critical-path capabilities
- Structured transition planning across phases
- Evidence-based policy and investment guidance

The methodology combines a pipeline-oriented transformation model with a multi-dimensional analytical framework, ensuring both structural traceability and depth of evaluation.

5.1 End-to-End Methodological Flow - summary

The methodology follows a structured pipeline:

1. Capabilities are defined as atomic inputs mapped to the architectural framework
2. Capabilities are evaluated through the 4-dimensional filtering logic
3. Selected capabilities are aggregated into priorities
4. Priorities evolve across phases through controlled transitions
5. The executive summary synthesizes the outcomes into strategic guidance

This ensures a transparent, traceable, and reproducible approach to roadmap development.

5.2 Capabilities as Structured Inputs

Capabilities constitute the primary analytical units of the roadmap and are directly derived from the architectural framework (Fig. 1). Each capability represents a discrete, actionable function required to ensure cybersecurity and cyber-resilience across the Cloud–Edge–IoT continuum.

Each capability is defined along three axes:

- **Layer Mapping:** Association to a specific layer (device, edge, cloud, application)
- **Cross-Layer Scope:** Identification of dependencies across layers (e.g., identity, trust, orchestration)
- **Functional Intent:** The security objective it fulfills (e.g., confidentiality, integrity, availability, sovereignty)

Capabilities are treated as **atomic inputs** to the prioritization methodology and are evaluated independently before aggregation into higher-level priorities.

5.3 4-Dimensional Transition Filtering Logic – from capabilities to Priorities

To ensure the roadmap provides actionable guidance for European R&I funding (Horizon Europe) and strategic investment, capabilities are prioritized using a structured 4-filter logic. This objective is to identify the “Critical Path” items that require immediate resource allocation during Phase 1 (0–5 years) in order to enable a secure and sovereign European Cloud–Edge–IoT computing continuum.

To determine which capabilities advance across implementation phases, a structured filtering logic is applied. This logic acts as a **decision function** that evaluates each capability against four dimensions:

1. **Regulatory Urgency – Compliance Filter (The “Must-Do” Filter):** Alignment with imminent EU regulatory requirements (e.g., NIS2, CRA, AI Act)

- **Question:** Is this capability required to meet imminent European regulatory obligations (e.g., NIS2 [1], Cyber Resilience Act [2], AI Act [3])?
- **Logic:** If failure to achieve this by 2026 results in market exclusion or systemic non-compliance, it is designated as **Priority #1**.

2. **Foundational Dependency – Architecture Filter (The “Blocker” Filter):** Degree to which the capability acts as a prerequisite for other capabilities

- **Question:** Do subsequent security layers depend on this capability existing first?
- **Logic:** Foundational technologies (e.g., Layer 1 Hardware Trust) outrank high-level software orchestration. You cannot secure the user (Layer 4) if the hardware (Layer 1) cannot provide a verifiable anchor.

3. **The Sovereignty Impact / Strategic Sovereignty Risk – Resilience Filter (The “Extinction” Filter):** Contribution to European digital sovereignty and strategic autonomy

- **Question:** Does a lack of this tech pose an existential threat to EU sovereignty or systemic resilience?
- **Logic:** This prioritizes areas where the EU currently has 100% foreign dependency or where a technical breakthrough (e.g., *Q-Day* Quantum threat) would render existing defenses obsolete.

4. **Market Feasibility & Ecosystem Alignment – Adoption Filter (The “Simpl/SME” Filter):** Technical maturity, implementation readiness, and deployment constraints

- **Question:** Does this align with existing EU heavy-investments (e.g., Simpl [6]) and is it adoptable by SMEs?
- **Logic:** Priority is given to capabilities that act as force-multipliers for existing infrastructure, ensuring economic viability.

Each dimension is evaluated qualitatively (High, Medium, Low) and used to determine whether a capability:

- Advances to the next phase
- Remains in the current phase
- Is deprioritized or deferred

Capabilities that satisfy multiple filters are classified as *Executive Priorities* and define the critical investment path for the secure European computing continuum.

Transition Definition and Logic

Within this methodology, a **transition** represents the structured transformation of outputs between stages of the pipeline. Transitions are governed by explicit mapping rules rather than narrative interpretation:

- **Capability → Priority Transition (Stage 1 → Stage 2):** Driven by the 4-filter logic, selecting and ranking capabilities.

- **Priority → Profile Transition (Stage 2 → Stage 3):** Each selected priority is expanded into a full 7-dimensional analytical structure.
- **Profile → Matrix Transition (Stage 3 → Stage 4):** Selected dimensions (Baseline, Activities, Enablers) are systematically mapped into matrix indicators (ODP, SSL, Operational State, Critical Enabler).
- **Matrix → Strategy Transition (Stage 4 → Stage 5):** Cross-priority aggregation identifies shared enablers, systemic risks, and phase-level strategic posture.

This formalization ensures that each stage is reproducible, auditable, and traceable.

5.4 The Anatomy of a Roadmap Priority: A Hybrid 7-Dimensional Analysis Template

To ensure the roadmap is actionable, realistic, and aligned with European strategic autonomy, each priority is analyzed through a hybrid 7-dimensional profiling framework that combines methodological consistency with contextual flexibility. The template is applied consistently across priorities, with the first four dimensions treated as mandatory and the remaining dimensions applied where relevant based on risk exposure, system criticality, and policy impact.

Priority Profiling Framework: Mandatory and Adaptive Dimensions

To ensure both analytical rigor and flexibility across heterogeneous priorities, the 7-dimensional profiling framework is structured into:

- **Mandatory Core Dimensions:** Applied to every priority to ensure comparability, traceability, and alignment with EU strategic objectives.
- **Adaptive Contextual Dimensions:** Applied selectively depending on the nature, maturity, and risk profile of each priority.

This hybrid approach ensures that all priorities are evaluated consistently, while allowing deeper analysis where complexity, uncertainty, or systemic risk requires it.

Dimension Applicability Logic

Not all priorities require equal depth across all seven dimensions. Therefore:

- The **first four dimensions are mandatory** and must be defined for every priority.
- The remaining three dimensions are **conditionally applied** based on:
 - System criticality
 - Exposure to systemic or adversarial risk
 - Policy and societal relevance

This ensures proportional analysis effort while maintaining methodological consistency.

1. Baseline & Target Objective (The Trajectory) - Q: Where do we stand today, and what exactly does success look like?

Description: Establishes the starting point of the priority today and defines the ultimate maturity state required by the end of the 15-year roadmap.

Evaluation Output: Defines the maturity gap and informs phase transition readiness.

- **Baseline (Where we are):** Grounds the roadmap in reality. It captures the state of the art, current Operational Deployment Phase (ODP), Strategic Sovereignty Level (SSL), industrial capacity, reliance on foreign vendors, and the technical or economic barriers preventing immediate adoption.
- **Target (Where we need to be):** Sets the baseline for EU-funded projects, defining the target ODP and SSL that must be achieved by the end of the roadmap.

2. Primary Drivers (The “Push” Factors) - Q: Why must we prioritize building this

Table 1: Priority profiling dimensions and applicability

Dimension	Type	Purpose
Baseline & Target Objective	Mandatory	Define maturity trajectory and target state
Primary Drivers	Mandatory	Establish urgency and justification
High Impact Activities	Mandatory	Identify key interventions
Catalysts & Enablers	Mandatory	Define EU support mechanisms
Disruptors & Risks	Adaptive	Identify systemic and adversarial threats
Failsafe Mechanisms & Resilience	Adaptive	Ensure resilience under failure
Transversal Meta-Layer	Adaptive	Align with EU policy and societal goals

capability right now? SQ: Why are we doing this?

Description: Outlines the external pressures—both from the evolving threat landscape and mandatory EU regulations—that make this technological evolution urgent and unavoidable.

Interpretation: Determines urgency and influences prioritization ranking.

- **Threat Drivers:** The rapid advancement of adversarial capabilities (e.g., offensive AI, quantum decryption, supply chain targeting).
- **Regulatory Drivers:** Imminent EU compliance mandates that require technical operationalization (e.g., NIS2, Cyber Resilience Act, AI Act).

3. **High Impact Activities - Q: Which specific actions provide the greatest leap in European sovereignty and security? SQ: Which should be the key initiatives?**

Description: Highlights the specific, prioritized technical interventions within this broader capability group that will yield the highest strategic return on investment.

Output: A shortlist of actionable R&I and deployment interventions.

4. **Catalysts & Enablers (The “Pull” Factors & EU Support) - Q: What specific EU**

funding, standards, and ecosystem support do we need? SQ: How do we get there & accelerate?

Description: Identifies the concrete mechanisms, funding vehicles, and regulatory standardizations required from the EU to accelerate market adoption and push the industry from the baseline to the target state.

Role in Methodology: Bridges priorities with EU instruments, ensuring implementation feasibility.

- **R&I & Funding Alignment:** Required Horizon Europe funding topics and targeted innovation instruments. Investment priorities must be explicitly synchronized with the European Cybersecurity Competence Centre (ECCC) Strategic Agenda [8] to ensure alignment with pan-European goals (e.g., PQC transition infrastructure, AI-driven security tools).
- **Regulatory & Standardization:** Alignment with standard-setting bodies (CEN/CENELEC) and certification frameworks (e.g., EUCS, EUCC).
- **Ecosystem:** Support for ecosystem building, industrial partnerships, and capacity-building initiatives (e.g., federated Cyber Ranges) to upskill the workforce and ensure advanced security solutions are eco-

nomically viable for SMEs.

5. **Disruptors & Systemic Risks (The Roadblocks) - Q: What structural vulnerabilities, market failures, or adversary breakthroughs could derail us? SQ: What can break the path?**

Description: Acknowledges that innovation is not linear by identifying severe technical and architectural risks that could actively derail deployment. It places primary emphasis on “Technical Cliffs”—moments where defensive capabilities are rendered obsolete by adversarial breakthroughs before countermeasures are fully deployed.

Trigger Condition: Applied when the priority is exposed to technological discontinuities or adversarial asymmetry.

- **Cryptographic Obsolescence & Migration Bottlenecks (The Primary Technical Risk):**

- *The “Q-Day” Cliff:* The risk that Cryptographically Relevant Quantum Computers (CRQC) become operational before the EU completes its migration to Post-Quantum Cryptography (PQC). This creates a “Harvest Now, Decrypt Later” vulnerability.
- *Migration Complexity:* The systemic failure to inventory and replace deep-rooted Public Key Infrastructure (PKI) in legacy OT/IoT environments.

- **Adversarial Breakthroughs:** The risk of threat actors weaponizing emerging technologies (e.g., AI-driven exploit generation, polymorphic malware) faster than defensive R&I can mitigate them.

- **Contextual Risks (Non-Technical / Optional):** Geopolitics & supply chain dependencies, market & technology risks (monopolization, fragmented standards).

6. **Failsafe Mechanisms & Resilience (The Safety Nets) - Q: If this technology is compromised, how does the system survive**

and gracefully degrade? SQ: How do we prevent failure?

Description: The technical core of “Cyber-Resilience.” It defines the engineering fallbacks required if a primary capability fails, is breached, or is disrupted by the systemic risks identified above.

Purpose: Ensures continuity of operations under failure or compromise scenarios.

What it entails: Designing for degraded modes of operation, architectural redundancies, autonomous isolation protocols, and implementation diversity to avoid vulnerable technological monocultures.

7. **Transversal Meta-Layer (The Policy Alignment) - Q: How does mastering this capability support Europe’s goals of sovereignty, sustainability, and trust? SQ: The broader policy context?**

Description: Elevates the technical capability back to the overarching political and societal goals of the European Union.

Strategic Role: Aligns technical priorities with EU-wide policy narratives and long-term societal impact.

- **Sovereignty & Autonomy:** Strengthening European strategic independence, reducing reliance on non-EU vendors.
- **Trust & Ethics:** Fostering deep societal trust by guaranteeing data privacy (Privacy-by-Design), algorithmic fairness, and transparency.
- **Sustainable Security (Green IT):** Minimizing the environmental impact of cyber defense (e.g., energy-efficient cryptographic algorithms, crypto-agility to reduce e-waste).

5.5 Phase Transition Mechanism - Progression Matrix: Sequence of Events (X-Axis)

The roadmap is structured across multiple time phases (e.g., short-term, mid-term, long-term). Each

Table 2: Mapping from 7-Dimensional Profile to Progression Matrix

Profile Dimension	Matrix Component
Baseline & Target Objective	ODP and SSL starting/target levels
High Impact Activities	Operational State evolution
Catalysts & Enablers	Critical Enabler
Disruptors & Risks	Phase risk considerations
Failsafe Mechanisms	Resilience assumptions in synthesis

phase represents a column in the prioritization matrix.

Transitions between phases are governed by the evolution of capabilities and priorities across these columns.

Instead of a strictly chronological timeline, the roadmap maps the sequence of critical events required to unlock maturity. This ensures that foundational capabilities are built before advanced autonomous systems.

- **Phase 1: Foundation & Compliance (Immediate Implementation):** Focuses on state-of-practice deployments, bridging immediate capability gaps, and addressing urgent regulatory obligations (NIS2, CRA).
- **Phase 2: Federation & Standardization (Scaling & Harmonization):** Focuses on deploying emerging technologies, establishing harmonized governance frameworks, and managing critical transition periods.
- **Phase 3: Autonomy & Leadership (Sovereign Innovation):** Focuses on highly speculative but structured foresight, including defense against quantum-capable adversaries and the achievement of fully sovereign, autonomous cloud-edge ecosystems.

For each phase:

- **Input:** Set of prioritized capabilities from the previous phase
- **Processing:** Application of the 4-dimensional filtering logic

- **Output:** Updated set of priorities for the next phase

A transition occurs when:

- A capability increases in maturity or feasibility
- Dependencies are resolved
- Regulatory pressure increases
- Strategic importance shifts

Each transition produces a **phase summary**, capturing:

- Newly introduced priorities
- Evolved priorities
- Deferred or dropped priorities

Matrix Cell Content Definition (Measuring Progression)

Rather than relying on subjective Technology Readiness Levels (TRL) or Manufacturing Readiness Levels (MRL) for a highly fragmented ecosystem, each cell in the progression matrix tracks maturity using objective strategic market and policy indicators:

- **Operational Deployment Phase (ODP):** Tracks the market reality and integration status (e.g., *Applied R&D* → *Fragmented Piloting* → *Standardized Integration* → *Ubiquitous Operation*).

- **Strategic Sovereignty Level (SSL):** Tracks European strategic autonomy and reliance on foreign vendors (e.g., *High Foreign Dependency* → *Emerging EU Ecosystem* → *Full Sovereign Autonomy*).
- **Operational State:** A concise snapshot characterizing the reality on the ground during this phase.
- **Critical Enabler:** The single most critical regulation, standard, or European R&I breakthrough unlocking the transition to the next phase.

5.6 From Priorities to Executive Summary: Final Strategic Synthesis

After priorities are profiled and their cross-phase progression is captured in the transition matrix, a final synthesis step is applied to convert detailed technical analysis into executive-level strategic guidance.

The synthesis is non-normative and evidence-driven, relying exclusively on outputs generated in previous stages rather than introducing new assumptions.

This synthesis does not re-score priorities. Instead, it aggregates the strongest signals emerging from the previous methodological steps and translates them into decision-oriented messages for policymakers, funders, and industrial stakeholders.

Inputs to the Synthesis Step

The final synthesis reads and consolidates the following inputs:

- The selected **Executive Priorities** emerging from the 4-filter logic
- The **mandatory and adaptive dimensions** captured in each priority profile
- The **cross-phase evolution** recorded in the progression matrices, including ODP, SSL, operational state, and critical enablers

Synthesis Logic

This step defines the **aggregation logic** of the methodology, transforming individual priority analyses into system-level strategic insights. The synthesis step identifies:

- **Common patterns** across priorities (e.g., repeated dependencies on hardware trust, PQC migration, interoperability, or compliance automation)
- **Cross-cutting bottlenecks** that appear in multiple priorities and therefore require coordinated EU action
- **Strategic inflection points** where delays in one capability would cascade into multiple downstream priorities
- **Investment signals** indicating where Horizon Europe, Digital Europe, ECCC, or standardization efforts should be concentrated

Output of the Synthesis Step

The output of this step is an executive summary that distills the roadmap into:

- A concise set of **critical strategic messages**
- A **prioritized investment narrative** for Phase 1 and beyond
- A **policy-aligned explanation** of why these priorities matter for sovereignty, resilience, and trust

This ensures that the executive summary is fully traceable to the analytical pipeline rather than functioning as a standalone narrative.

5.7 Executive Summary as Aggregated Output

The executive summary represents the final synthesis of the methodology and the output of the strategic synthesis step. It is not an independent narrative, but a structured aggregation of:

- High-priority capabilities (from filtering)
- Cross-phase evolution trends (from transitions)
- Strategic gaps and investment needs

It provides:

- A consolidated view of critical priorities
- Recommendations for R&I funding and policy alignment
- Identification of strategic dependencies and risks

This ensures full traceability from raw capabilities to strategic recommendations.

6 Capabilities (Architectural Framework)

As it is illustrated in Figure 1, the roadmap is structured across a cohesive architectural framework that spans four continuum environments (HPC, Cloud, Edge, Devices). The capabilities are organized into Six Security Layers (The Vertical Stack) and Four Transversal Cross-Layers (The Horizontal Enablers).

Part A: The Security Layers (Vertical Stack)

These layers define the specific technical security controls implemented across the computing infrastructure.

Layer 1: Hardware-Centric Security

1.1 Hardware Roots of Trust: Cryptographically verifiable supply chain trust (e.g., TPM attestation, secure enclaves) ensures physical device integrity. Success depends on driving European open-hardware designs (e.g., RISC-V) toward **EUCC 'High' assurance level certification** [5] to mathematically guarantee resistance against state-level physical tampering and side-channel attacks at the vulnerable far-edge. **1.2 Physical-Layer (PHY) Threat Detection & Fingerprinting:** Exploiting intrinsic wireless channel characteristics, such as Channel State Information (CSI), Carrier Frequency Offset (CFO), and Received Signal Strength Indicator (RSSI). That is to establish real-time, keyless device authentication and anomaly detection at the lowest layer of the stack. This provides a hardware-anchored security primitive that is resilient even when upper-layer cryptographic processing is compromised or infeasible.

Layer 2: Software-Centric Security (Middleware/OS)

2.1 OS & Middleware Hardening: Securing hypervisors, container runtimes, and distributed operating systems is critical for isolation. This involves hardening KVM/QEMU and supporting sovereign,

Security Framework with Separated Security Verticals

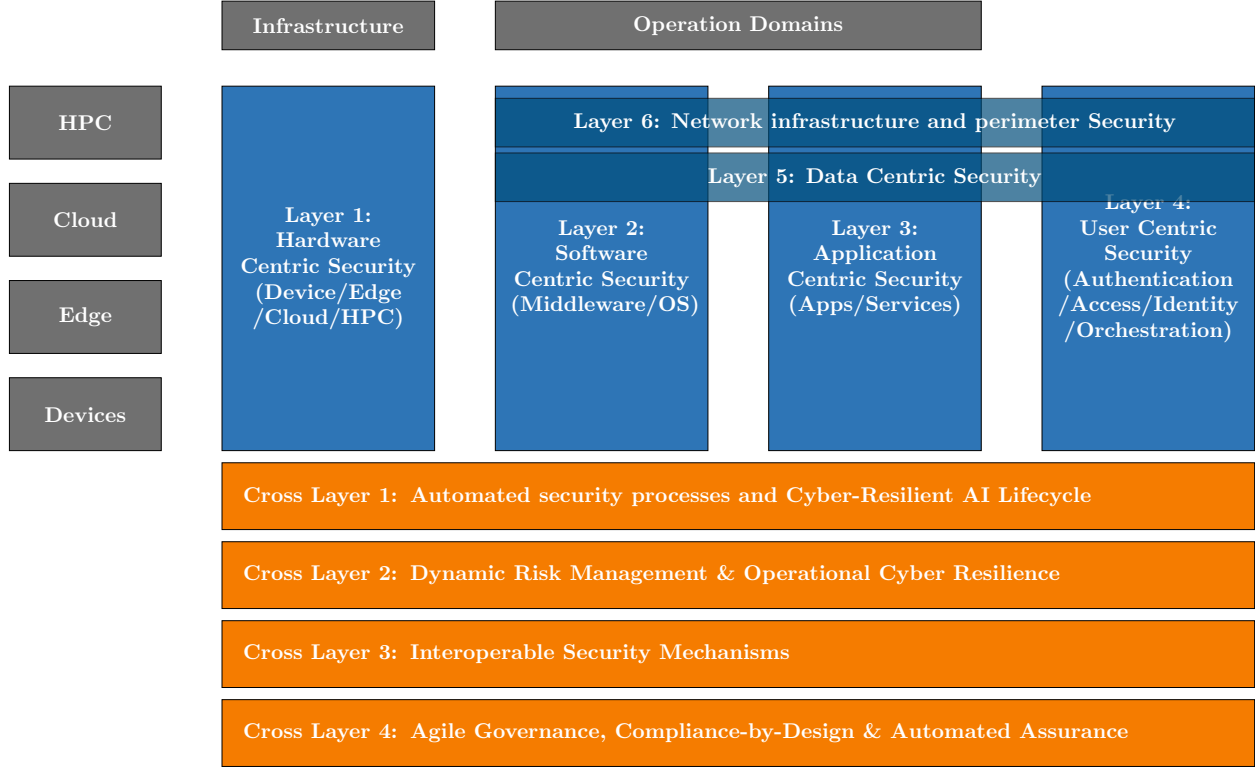


Figure 1: Security Framework with separated security verticals in dedicated boxes.

lightweight Linux distributions optimized for edge resource constraints, ensuring a reduced attack surface for the cognitive continuum.

2.2 Secure OTA Patch & Update Management: Edge nodes deployed in the field—smart grids, O-RAN radios, industrial gateways—require cryptographically verified, failure-resilient over-the-air (OTA) update pipelines. A compromised or failed update mechanism can render entire fleets of field-deployed nodes permanently vulnerable, making robust OTA management a critical software-layer capability.

Layer 3: Application-Centric Security (Apps/Services)

3.1 Workload Security: Securing containerized applications, microservices, and serverless functions executing dynamically across multi-provider nodes. This is achieved through hardened service meshes and sidecar proxy architectures that provide mutual TLS and granular traffic inspection without impacting performance.

3.2 In-Band Network Telemetry (INT) & Deep Observability: Embedding “smart telemetry” directly into the data packets traversing containerized workloads and service meshes, providing

granular, real-time visibility into network behaviour and performance without requiring out-of-band monitoring infrastructure. This feeds high-fidelity signals to security analytics platforms.

3.3 API Security Gateway & Policy Enforcement: As containerized AI workloads increasingly communicate through REST/gRPC APIs, a dedicated API security gateway provides a unified enforcement point for authentication tokens, rate limiting, schema validation, and payload inspection—preventing injection attacks and data exfiltration at the application boundary.

Layer 4: User & Orchestration-Centric Security

4.1 Zero-Trust Orchestration: Seamless policy enforcement across a multi-provider computing continuum requires robust Policy Enforcement Point (PEP) and Policy Decision Point (PDP) architectures. This enables continuous authorization and dynamic access controls that adapt to the context of the request.

4.2 Federated Identity & Access Management (IAM): Cross-domain identity federation and verifiable credentials for machines, users, and automated workloads. Implementing **Self-Sovereign Identity (SSI)** principles allows for cryptographically secure, decentralized authentication across heterogeneous cloud-edge domains.

4.3 Privileged Access Management (PAM) & Just-In-Time (JIT) Access: Lateral movement by attackers frequently exploits over-provisioned standing privileges—the precise vector behind major Advanced Persistent Threat (APT) campaigns. PAM and JIT access controls ensure that elevated privileges are granted only for defined, time-bounded sessions with full auditability, minimizing the blast radius of any credential compromise.

Layer 5: Data-Centric Security

5.1 End-to-end Confidential Computing: Data-in-use security through Trusted Execution Environments (TEEs) spanning the entire Cloud-to-IoT data path. Leveraging TEE abstraction layers and re-

mote attestation ensures that sensitive European data remains encrypted even during processing in non-trusted environments.

5.2 Post-Quantum & Cryptographic Agility: PQC standardization and hybrid crypto deployment are necessary to secure long-lived data. Priority is placed on **PQC migration for resource-constrained devices**, ensuring that edge infrastructure is resilient against “Harvest Now, Decrypt Later” quantum threats.

5.3 Signaling (Data) Security: Moving beyond best-effort packet forwarding by embedding trust directly into the network signaling architecture. This ensures the integrity and authenticity of control-plane messages, preventing hijacking of routing decisions and network configuration across the distributed continuum.

5.4 Distributed Reputation & Trust Networking: Implementing a ubiquitous, real-time reputation system built on Distributed Ledger Technology (DLT) and smart contracts. This enables autonomous, evidence-based trust decisions about nodes, services, and data sources across the continuum without reliance on a central authority.

5.5 Quantum Key Distribution (QKD) for Critical Links: While PQC addresses algorithmic quantum-resistance, QKD provides information-theoretically secure key exchange over optical fibre for point-to-point high-value links (government, critical infrastructure backbone). These two technologies are complementary and together form a comprehensive quantum-safe strategy.

5.6 Secure Multi-Party Computation (SMPC): Homomorphic Encryption and TEEs (Layer 5.1) handle confidential computing for single-party workloads. SMPC enables multiple mutually distrusting parties to jointly compute a function over their combined private inputs without revealing those inputs to each other—a critical capability for cross-border analytics and federated AI in regulated sectors.

Layer 6: Network Infrastructure & Perimeter-Centric Security

6.1 Telco-cloud & O-RAN Security: Hardening distributed 5G/6G cores and RAN components must align with the **Smart Networks and Services Joint Undertaking (SNS JU) SRIA** [7]. This ensures sovereign control over network slice isolation and virtualized RAN components.

6.2 Cross-Sector Physical Resilience & Safety: Adapting strict network controls and IT/OT gateway security for high-availability edge environments (e.g., Smart Grids). This guarantees that cyber-breaches in the digital layer do not cascade into physical safety hazards or kinetic damage in critical infrastructure.

6.3 Edge-Native Autonomous Guardians: Hosting lightweight, localized security control functions (Security-as-a-Service aggregators) directly at the far-edge gateways or within Open-RAN Radio Intelligent Controllers (RIC). These autonomous guardian nodes execute real-time threat mitigation without requiring round-trip communication to a centralized cloud-based SOC, dramatically reducing response latency.

6.4 Non-Terrestrial Network (NTN) & Satellite Security: 6G will natively integrate satellite segments for coverage continuity, backhaul resilience, and emergency operations. These non-terrestrial links introduce distinct threat surfaces—including signal interception and orbital spoofing—requiring dedicated security requirements aligned with 3GPP NTN specifications and the EU Space Programme.

Part B: The Transversal Security Verticals (Cross-Layers)

These four cross-layers represent continuous, automated processes and governance structures that intersect every layer of the vertical stack.

Cross-Layer 1: Automated Security Processes & Cyber-Resilient AI Lifecycle

CL1.1 Continuous DevSecOps (Shift-Left): Embedding AI-driven code analysis and automated vulnerability remediation at the earliest stages of the software lifecycle.

CL1.2 Model & Pipeline Integrity (DataBOMs): Securing AI training pipelines to prevent data poisoning and unauthorized model manipulation.

CL1.3 Runtime AI Defense: Defending AI/ML infrastructure against real-world adversarial attacks (e.g., prompt injection, model evasion/theft) at the edge.

CL1.4 Dynamic Supply Chain Verification (Runtime SBOMs): Moving beyond static software manifests by implementing dynamic Software Bill of Materials (Runtime SBOMs) and “cryptographic proof of execution” to continuously verify the integrity of containerized microservices and AI components as they migrate across the multi-provider edge-cloud continuum.

CL1.5 Automated Adversarial Testing & Continuous Purple Teaming: DevSecOps pipelines focus on code analysis and remediation; the adversarial counterpart provides continuous automated penetration testing, fuzzing (coverage-guided and protocol-level), and purple-team automation that simulates real attacker behaviour against live staging environments, ensuring defenses are validated continuously rather than at periodic intervals.

Cross-Layer 2: Dynamic Risk Management & Operational Cyber Resilience

CL2.1 Automated SOC Operations & Collaborative Threat Intelligence: Continuous monitoring, SOC federation across Member States, and automated incident containment, underpinned by legally protected, cross-border threat intelligence sharing ecosystems designed to overcome institutional trust barriers and incentivize active collaboration [9].

CL2.2 Systemic & Cascading Risk Mitigation: Mapping the evolving threat landscape and using impact-likelihood scoring to dynamically sever

links and prevent cross-domain systemic failures.

CL2.3 High-Precision Telemetry and Security Signalling: Implementing “smart telemetry” and cybersecurity-related signalling to feed real-time threat intelligence into analytics platforms without overwhelming network bandwidth—ensuring that visibility does not itself become an attack vector or performance bottleneck.

CL2.4 Internet of Behavior (IoB) and Evidence Sharing: Utilizing cybersecurity and QoS signalling traffic to extract and share behavioral patterns across the infrastructure, establishing an “Internet of Behavior” that enables proactive anomaly correlation and evidence-based risk assessment at continental scale.

CL2.5 AI-Native Analytics Integration (NWDAF & LLMSecOps): Routing cybersecurity signalling traffic and telemetry through centralized analytics hubs powered by AI agents. This creates a live “nervous system” for the network, allowing AI agents to continuously monitor for zero-day exploits, analyze encrypted traffic flows, and trigger automated incident responses aligned with NWDAF (Network Data Analytics Function) standards.

CL2.6 Cyber Threat Intelligence (CTI) Lifecycle Management: While CL2.1 and CL2.4 consume and act on threat data, explicit governance of the full CTI lifecycle—collection, processing, analysis, dissemination, and feedback—is required as a prerequisite for the federated SOC vision. A sovereign EU CTI platform must define this governance layer to ensure data quality, legal compliance, and attribution reliability.

CL2.7 Cyber Deception Technology & Active Defense (Honeynets, Canaries): Current detection is passive—telemetry is observed and anomalies flagged. Deception adds an active intelligence-collection dimension by deploying decoy assets (honeypots, honeynets, credential canaries, fake API endpoints) that generate high-fidelity, low-noise alerts when touched, enabling early detection of advanced intrusions and insider threats.

Cross-Layer 3: Interoperable Security Mechanisms & Open-Source Verifiability

CL3.1 Heterogeneous Interoperability & Simpl Integration: Secure communication and data exchange across disparate systems, edge nodes, and proprietary domains. This includes mandatory alignment with and integration into the Simpl open-source smart middleware [6] to ensure secure, federated access across Common European Data Spaces.

CL3.2 Minimal Interoperability Mechanisms (MIMs) & Open Verifiability: Establishing standardized, baseline security and privacy controls to ensure trust without creating vendor lock-in. This relies heavily on embracing open-source software and open hardware architectures as fundamental prerequisites for transparency, cryptographic verifiability, and the elimination of proprietary black-box vulnerabilities [6].

CL3.3 Neutral Host and Multi-Tenant Isolation: As the ecosystem shifts toward infrastructure sharing and Neutral Host models spanning wireless, optical, and cloud domains, robust isolation between tenants is critical. Security architectures must enforce Zero-Trust edge operations across different network partitions to prevent cross-tenant data leakage and privilege escalation.

CL3.4 Pan-European NaaS Federation and Cross-Border Edge Roaming: Establishing standardized interfaces and harmonized Network APIs (e.g., leveraging CAMARA) to interconnect the fragmented edge clouds of different European telecom operators into a unified Network-as-a-Service (NaaS) platform—enabling a “Schengen for Cloud/AI” where compute workloads can roam seamlessly across borders.

CL3.5 Semantic Security Interoperability (Common Ontologies & Data Models): Federated SOC and shared threat intelligence only function if participating organisations share a common semantic understanding of security events, asset taxonomies, and risk classifications. Standardized ontologies (e.g., aligned with STIX/TAXII and MITRE ATT&CK) are a prerequisite for meaningful cross-border correlation.

CL3.6 Cyber Range & Digital Twin Federation for Security Validation: Before deploying cross-border federated security configurations into production, they must be validated in high-fidelity simulation environments. ENISA already coordinates national cyber range inventories; this cross-layer would federate them into a shared EU-wide infrastructure aligned with the Cyber Solidarity Act’s testing provisions.

Cross-Layer 4: Compliance-by-Design, Agile Governance & Automated Assurance

CL4.1 Policy-as-Code & Compliance-by-Design: Technical implementation of legal requirements (OSCAL) into machine-readable rules that pre-validate workloads.

CL4.2 Continuous Automated Auditing & Certification Lifecycle: Replacing “point-in-time” PDF certificates with real-time, telemetry-driven certification tracking for both cloud services (EUCS) [4] and physical ICT edge products (EUCC) [5]. This ensures that edge nodes autonomously maintain compliance by integrating mandatory vulnerability handling and cryptographic patch verification directly into their operational lifecycle, strictly aligning with EUCC maintenance and assurance rules [5].

CL4.3 Post-Quantum Readiness & Zero-Trust Architectures (ZTA): Moving beyond traditional perimeter-based defenses by embedding continuous, AI-driven Zero-Trust verification and micro-segmentation natively into the cloud-edge infrastructure, combined with a systematic migration roadmap to post-quantum cryptographic primitives across all layers.

CL4.4 Continuous Automated Certification: Replacing “point-in-time” PDF certificates with real-time, telemetry-driven certification tracking for both cloud services (EUCS) and physical ICT edge products (EUCC), ensuring that edge nodes autonomously maintain compliance by integrating mandatory vulnerability handling and cryptographic patch verification directly into their operational lifecycle [5].

CL4.5 Multicloud Key Management as a Service (KMaaS): When AI workloads, data flows,

and context flows migrate dynamically across a multi-provider computing continuum, managing cryptographic keys and access policies becomes highly complex. KMaaS provides a provider-agnostic, sovereign key lifecycle management capability that prevents key sprawl and ensures revocation propagates consistently across heterogeneous environments.

CL4.6 Privacy-Preserving Federated Computation: When multiple providers work on the same infrastructure, AI training and inference may require access to datasets stored across different jurisdictions and competitive boundaries. Privacy-preserving techniques (federated learning, differential privacy, SMPC) enable collaborative computation while guaranteeing that raw data never leaves its sovereign jurisdiction.

CL4.7 Cyber Risk Quantification & Board-Level Reporting (CRQ): Compliance frameworks define what controls to implement. A dedicated cross-layer translates technical security posture into financial and operational risk language for decision-makers—potentially in real-time—enabling boards to make informed, evidence-based investment and risk-acceptance decisions aligned with DORA requirements.

CL4.8 AI Trustworthiness & Algorithmic Accountability: As AI systems themselves become actors within the security architecture—making autonomous containment decisions, classifying threats, orchestrating responses—the trustworthiness of those AI systems becomes a security-critical concern. This layer defines explainability, auditability, and override mechanisms for AI-driven security functions, ensuring alignment with the EU AI Act for high-risk systems.

CL4.9 Regulatory Digital Twins: Virtualized replicas of critical infrastructure specifically to stress-test regulatory compliance against evolving laws before applying changes to the live physical continuum.

7 Priorities

7.1 Executive Priorities: The Critical Path (Phase 1)

Applying the 4-Filter Framework, the following areas are identified as immediate strategic priorities for the 2025-2030 funding cycle:

- **Priority 1:** Hardware Roots of Trust & EUCC 'High' Certification (Filters 1, 2, 3).
- **Priority 2:** Post-Quantum Cryptographic Agility for Edge/IoT (Filter 3).
- **Priority 3:** Lightweight Cryptography & Physical-Layer Security (PhySec) for the Extreme Edge.
- **Priority 4:** Security Interoperability via Simpl Middleware (Filter 4).
- **Priority 5:** Supply Chain Verification and Secured Infrastructure.
- **Priority 6:** Cognitive SecOps (LLMSecOps) & Automated Threat Intelligence.
- **Priority 7:** Pan-European NaaS Federation ("Schengen for Cloud-Edge/AI").
- **Priority 8:** Confidential Computing & Privacy-Preserving AI Pipelines.
- **Priority 9:** Resilient-by-Design Infrastructure & Autonomous "Island Mode."
- **Priority 10:** Proactive Cyber Threat Management & Defense against Advanced Persistent Threats (APTs).

8 Capability Profiles & Progression Matrices

Priority 1: Hardware Roots of Trust & EUCC 'High' Certification

1. **Baseline & Target Objective:** *Baseline:* High foreign dependency on proprietary Trusted

Platform Modules (TPMs) and secure elements. Far-edge devices lack standardized hardware anchors. *Target:* Ubiquitous deployment of EU-designed, open-source (RISC-V) secure enclaves certified at **EUCC 'High' assurance**, providing a mathematically verifiable foundation for the entire continuum.

2. **Primary Drivers:** *Threat:* State-level physical tampering and side-channel attacks on decentralized edge nodes. *Regulatory:* Mandatory hardware security requirements under the **Cyber Resilience Act (CRA)** and certified anchors for **NIS2** critical infrastructure.
3. **High Impact Activities:** Development of sovereign European Secure Elements based on RISC-V; automated hardware-level remote attestation protocols for large-scale edge fleets; EU-level standard toolset developed and maintained by the European Commission; SME support for certification and compliance testing; security-based sensor research for future deployment.
4. **Catalysts & Enablers:** **EU Chips Act** pilot lines for secure silicon; **ECCC Strategic Agenda** funding for open-hardware security; alignment with CEN/CENELEC hardware security standards.
5. **Disruptors & Systemic Risks:** Supply chain bottlenecks for EU-based fabrication; breakthroughs in physical side-channel analysis that bypass current cryptographic enclaves.
6. **Failsafe Mechanisms & Resilience:** "Defense-in-depth" where software-based isolation (TEEs) acts as a secondary layer if a hardware enclave is compromised; autonomous node isolation and distributed attestation if hardware integrity is lost.
7. **Transversal Meta-Layer:** *Sovereignty:* Eliminates "black-box" hardware dependencies on non-EU vendors. *Sustainability:* Energy-efficient secure silicon designs reducing the carbon footprint of cryptographic operations at the far-edge.

Progression Matrix: Hardware Roots of Trust & EUCC 'High'

Priority 2: Post-Quantum Cryptographic (PQC) Agility for Edge/IoT

1. **Baseline & Target Objective:** *Baseline:* Current IoT and edge infrastructure predominantly relies on lightweight cryptography primitives which can be vulnerable to future quantum computers. Limited awareness and no systematic migration path for resource-constrained devices. *Target:* A “Quantum-Native” continuum where all critical edge/IoT infrastructure has migrated to hybrid or full Post-Quantum Cryptography, with automated crypto-agility enabling seamless algorithm updates without hardware replacement.
2. **Primary Drivers:** *Threat:* The Q-Day threat and “Harvest Now, Decrypt Later” strategies targeting sensitive long-lived data. *Regulatory:* Emerging ENISA guidelines and future EU mandates requiring PQC readiness for critical infrastructure under NIS2.
3. **High Impact Activities:** Optimization of NIST-standardized PQC algorithms for 8/16-bit microcontrollers; development of crypto-agility middleware for OTA algorithm updates; automated cryptographic inventory and migration tools for legacy IoT deployments; 3GPP specifications alignment for quantum key distribution (QKD) in cellular systems.
4. **Catalysts & Enablers:** Post-Quantum Cryptography standardization initiatives; **3GPP** specifications for quantum key distribution (QKD); **NIST PQC standardization finalization**; **ETSI** quantum-safe cryptography standards; **ENISA PQC guidelines**; **ECCC Strategic Agenda** funding; EU Chips Act for PQC-native silicon development.
5. **Disruptors & Systemic Risks:** Early arrival of CRQC before migration is complete; performance overhead of PQC algorithms on

battery-powered IoT; fragmented standardization; insufficient cryptographic inventory of legacy OT/IoT systems.

6. **Failsafe Mechanisms & Resilience:** Deployment of hybrid cryptographic schemes (classical + PQC) during the transition; crypto-agility frameworks allowing rapid algorithm switching if vulnerabilities are discovered; diversification of PQC algorithm families to avoid single-point cryptanalytic failures.
7. **Transversal Meta-Layer:** *Sovereignty:* Protection of Europe’s long-term industrial and governmental secrets from future quantum decryption. *Sustainability:* Energy-efficient, lightweight PQC implementations reducing the carbon footprint of security (Green IT).

Progression Matrix: PQC Agility for Edge/IoT

Priority 3: Lightweight Cryptography & Physical-Layer Security (PhySec) for the Extreme Edge

1. **Baseline & Target Objective:** *Baseline:* Massive IoT and nano-devices cannot support computationally heavy classical asymmetric encryption. Current security approaches are unsuitable for environments requiring 100 devices per m³ with 40-year battery lifespans or zero-energy backscatter radios. *Target:* Move to ultra-lightweight Post-Quantum Cryptography and keyless Physical-Layer Security (PhySec) embedded at the extreme edge, enabling security primitives that operate within severe energy and compute constraints.
2. **Primary Drivers:** *Threat:* The proliferation of massive Machine Type Communication (mMTC) creates an enormous unprotected attack surface; classical cryptography is both computationally infeasible and imminently vulnerable to quantum algorithms at the extreme edge. *Regulatory:* CRA requirements extend to the

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Fragmented Piloting	Standardized Integration	Ubiquitous Operation
SSL	High Foreign Dependency	Emerging EU Ecosystem	Full Sovereign Autonomy
Operational State	Cloud-based TPMs common; Edge nodes lack hardware trust; RISC-V pilots begin; CRA compliance starts.	Standardized EU-made secure elements in critical infra; Automated attestation at scale.	Sovereign hardware trust embedded by default in all EU ICT products from IoT to HPC.
Critical Enabler	Cyber Resilience Act (CRA) enforcement	EUCC 'High' certification of EU-based secure elements	Maturity of EU Chips Act sovereign fabrication lines

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Applied R&D	Fragmented Piloting	Ubiquitous Operation
SSL	High Foreign Dependency	Emerging EU Ecosystem	Full Sovereign Autonomy
Operational State	NIST algorithm selection; software-based PQC pilots in cloud environments; research on lightweight implementations.	Hybrid (classical + PQC) schemes in Industrial IoT and critical edge infrastructure; EU-developed crypto-agility middleware.	Fully quantum-resilient continuum with PQC embedded in all devices; automated crypto-agility across Cloud-Edge-IoT.
Critical Enabler	ENISA PQC Guidelines and NIST standardization completion	NIST/ETSI algorithm finalization and EU-wide deployment standards	PQC-native RISC-V silicon and ubiquitous crypto-agility frameworks

smallest connected devices; NIS2 mandates security across the full infrastructure including resource-constrained nodes.

- High Impact Activities:** Group-based authentication for large-scale constrained IoT fleets; RF fingerprinting and Channel State Information (CSI)-based device authentication; integration of lightweight PQC into European open-hardware (RISC-V) for smart body-area networks (WBAN); ETSI SmartBAN standardization alignment; research into “zero-energy” air interfaces with integrated Integrated Sensing and Communication (ISAC) security.
- Catalysts & Enablers:** NIST lightweight post-quantum cryptography standardization; ETSI SmartBAN standardization; 6G research into RF fingerprinting and channel entropy for native security; Hardware-Software Co-Design for neuromorphic and Processing-in-Memory (PIM) architectures.

- Disruptors & Systemic Risks:** Performance gap between available energy budgets and minimum cryptographic processing requirements; fragmented device ecosystems preventing uniform PhySec adoption; lack of EU-manufactured ultra-constrained secure silicon.
- Failsafe Mechanisms & Resilience:** Dynamic fallback to localized, non-routed trust boundaries leveraging purely physical-layer signatures (RF fingerprinting or channel entropy) to authenticate devices if upper-layer cryptographic processing exceeds the available energy budget or is compromised.
- Transversal Meta-Layer:** *Sovereignty:* Sovereign control over the security of the most proliferated and currently most unprotected layer of the computing continuum. *Sustainability:* Zero-energy security primitives are inherently aligned with Green IT objectives and circular economy principles.

Progression Matrix: Lightweight Cryptography & PhySec for the Extreme Edge

Priority 4: Security Interoperability via Simpl Middleware

1. **Baseline & Target Objective:** *Baseline:* Current European cloud and edge infrastructure is highly fragmented across proprietary vendor silos with incompatible security policies, authentication mechanisms, and data exchange protocols. This fragmentation prevents seamless data flow across Common European Data Spaces and creates vendor lock-in. *Target:* A unified European “Trust Fabric” enabling secure, federated interoperability across heterogeneous cloud and edge providers through mandatory integration with the **Simpl** open-source smart middleware stack, with standardized security Minimal Interoperability Mechanisms (MIMs).
2. **Primary Drivers:** *Regulatory:* **Data Act** mandates requiring seamless data portability and interoperability; requirements for Common European Data Spaces to operate across multiple providers; NIS2 requirements for coordinated incident response. *Market:* Breaking vendor lock-in to enable European SMEs to compete; enabling cross-border data flows while maintaining sovereignty.
3. **High Impact Activities:** Implementation and standardization of security-specific MIMs within the Simpl stack; automated policy translation engines converting provider-specific security policies into common machine-readable format (XACML, OSCAL); federated identity bridges using verifiable credentials; cross-domain security orchestration for unified threat response.
4. **Catalysts & Enablers:** **Digital Europe Programme** funding; **Simpl** open-source stack community and governance; **Gaia-X** federation services and trust frameworks; ECCC support for security MIM standardization.

5. **Disruptors & Systemic Risks:** Vendor resistance to open standards; weakest-link security failures in federated systems; complexity of policy translation creating semantic gaps; fragmented national implementations creating “data islands.”

6. **Failsafe Mechanisms & Resilience:** Capability to autonomously “unfederate” or isolate nodes that fail to meet the shared security baseline; zero-trust architectures where federation does not imply implicit trust; continuous compliance monitoring with automated ejection of non-compliant providers.
7. **Transversal Meta-Layer:** *Sovereignty:* Breaking dependence on non-EU hyperscaler lock-in. *Trust:* Fostering pan-European cross-border trust through transparent, verifiable security mechanisms. *Economic Viability:* Enabling SMEs to participate in Data Spaces without massive infrastructure investments.

Progression Matrix: Security Interoperability via Simpl Middleware

Priority 5: Supply Chain Verification and Secured Infrastructure

1. **Baseline & Target Objective:** *Baseline:* Supply chain tracking is static, opaque, and fragmented. The disaggregation of telco clouds (e.g., Open RAN) introduces extreme complexity by mixing software and hardware from multiple vendors. SMEs lack resources to meet CRA and NIS2 supply chain compliance requirements. *Target:* Move to continuously verified ecosystems utilizing Runtime Software Bill of Materials (SBOMs), Digital Product Passports (DPP), and automated certification across multi-vendor deployments, with full supply chain transparency and sovereign control.
2. **Primary Drivers:** *Threat:* Sophisticated supply chain attacks (e.g., SolarWinds-style compromises) targeting the disaggregated Open

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Isolated Lightweight Piloting	Standardized D2D Security Integration	Ubiquitous Zero-Energy Trust
SSL	High Dependency on Legacy/Heavy Algorithms	Emerging EU Lightweight & PhySec Ecosystem	Full Sovereign Extreme-Edge Cryptography
Operational State	Group-based authentication and early lightweight PQC algorithms tested on resource-constrained IoT; basic RF fingerprinting used for anomaly detection.	Lightweight PQC and PhySec natively embedded in European open-hardware (RISC-V) and smart body-area networks (WBAN).	Battery-less (zero-energy) nodes autonomously execute quantum-resistant authentication and trusted decentralized swarm communications.
Critical Enabler	NIST Lightweight PQC Standards & Context-Aware IoT Platforms	Hardware-Software Co-Design for Neuromorphic/PIM AI	“Zero-Energy” Air Interfaces & Ubiquitous ISAC (Integrated Sensing and Communication)

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Fragmented Piloting	Standardized Integration	Ubiquitous Operation
SSL	Emerging EU Ecosystem	Full Sovereign Autonomy	Full Sovereign Autonomy
Operational State	Initial Simpl security pilots in select Data Spaces; manual policy coordination; proof-of-concept MIM implementations.	Automated policy translation via standardized MIMs; Simpl-based security in majority of Common Data Spaces; cross-border federated SOC operations.	Fully automated, provider-agnostic European trust fabric; real-time security orchestration across the continuum; zero-trust federation by default.
Critical Enabler	Simpl-Open official launch and initial MIM definitions	Security MIMs adoption as CEN/CENELEC standards	EU Cloud Rulebook maturity and Data Act full enforcement

RAN and cloud-edge ecosystem; hidden backdoors in non-EU hardware and firmware. *Regulatory:* Strict legal mandates from the **Cyber Resilience Act (CRA)** and **NIS2 Directive** demanding robust supply chain risk management; Digital Product Passport (DPP) initiatives for traceability.

3. **High Impact Activities:** Review and mapping of hardware, OS, and software supply chain vulnerabilities; SME engagement prioritized—ensuring one-size-fits-all approaches are avoided and SMEs can enable new technology integration (e.g., NXP ecosystems); Dynamic Runtime SBOMs tracking active component states;

Digital Product Passports for end-to-end traceability; Zero-Knowledge Proofs (ZKPs) for IP-protected component verification.

4. **Catalysts & Enablers:** Digital Product Passport (DPP) initiatives; European financial support for SME security upskilling; ECCC strategic funding for supply chain security tooling; Pan-European threat intelligence sharing for supply chain indicators of compromise.
5. **Disruptors & Systemic Risks:** Extreme complexity of multi-vendor Open RAN supply chains; geopolitical dependencies on non-EU component manufacturers; SMEs being priced

out of compliance; lack of standardized machine-readable supply chain attestation formats.

6. **Failsafe Mechanisms & Resilience:** Graceful isolation of unverified or high-risk components into quarantined network slices until cryptographic proofs of execution can validate their integrity; automated blocking of unverified components in production deployments.
7. **Transversal Meta-Layer:** *Sovereignty:* Reducing systemic dependency on non-EU hardware and software supply chains. *Trust:* Providing verifiable, transparent provenance for all components in the European computing continuum. *Economic:* Protecting EU industrial IP through privacy-preserving verification mechanisms.

Progression Matrix: Supply Chain Verification and Secured Infrastructure

Priority 6: Cognitive SecOps (LLMSecOps) & Automated Threat Intelligence

1. **Baseline & Target Objective:** *Baseline:* Security Operations Centers (SOCs) are predominantly manual, reactive, and siloed. Analyst capacity is far outpaced by the expanding attack surface of 5G/6G and the millions of IoT devices. EU SOCs are heavily reliant on non-EU AI foundation models. *Target:* Autonomous, AI-driven “LLMSecOps” agents distributed across the 6G edge, capable of proactive threat hunting, real-time incident containment, and continuous learning without requiring human intervention for routine responses.
2. **Primary Drivers:** *Threat:* The dual-use nature of AI generating complex offensive threats (APTs, data poisoning, AI-driven exploit generation); the expanding attack surface of 6G and millions of IoT devices overwhelming human-scale SOC operations. *Regulatory:* EU AI Act compliance for high-risk autonomous security

systems; Zero-Touch Network & Service Management (ZSM) goals; EuroHPC AI Factories as the compute substrate for sovereign AI models.

3. **High Impact Activities:** LLM-assisted human SOC analysts integrated with NWDAF telemetry; edge-native “Guardian” nodes executing Zero-Trust reconfigurations autonomously; European sovereign Small Language Models (SLMs) for security classification; swarm-based agentic AI for decentralized threat intelligence; standardized AI Interconnect APIs.
4. **Catalysts & Enablers:** **EU AI Act** compliance frameworks for high-risk security AI; **EuroHPC AI Factories** providing compute for sovereign foundation model training; **ECCC Strategic Agenda** for LLMSecOps standardization; NWDAF integration with 3GPP standards.
5. **Disruptors & Systemic Risks:** AI hallucination causing false containment actions disrupting critical services; adversarial attacks on the AI models themselves (prompt injection, model poisoning); regulatory uncertainty around autonomous AI decision-making in security contexts; continued dominance of non-EU foundation models.
6. **Failsafe Mechanisms & Resilience:** “Shadow-mode” rollback and human-in-the-loop (RLHF) intervention if autonomous edge agents hallucinate or exhibit unstable control loops; mandatory override mechanisms for all autonomous security decisions affecting critical infrastructure.
7. **Transversal Meta-Layer:** *Sovereignty:* Development of European sovereign edge LLMs and SLMs replacing dependency on US foundation models. *Trust:* Ensuring that AI-driven security decisions are explainable, auditable, and aligned with the EU AI Act for high-risk systems. *Sustainability:* Energy-efficient edge AI inference reducing the carbon footprint of autonomous security operations.

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Full Supply Chain Disclosure	Partial Restriction Process	Full Restriction Process
SSL	High Reliance on Fragmented, Non-EU Supply Chain Auditing	Emerging EU Supply Chain Tracking Frameworks (DPP)	Full Sovereign Control and Supply Chain Transparency
Operational State	Review of hardware, OS, and software supply chain vulnerabilities; initial Runtime SBOM pilots; CRA compliance mapping begins.	Ensuring certification of critical supply chain hardware, OS, and software; DPP frameworks operational; SME compliance pathways established.	Fully certified supply chain actively blocking unverified components; ZKPs enable IP-protected verification at continental scale.
Critical Enabler	CRA supply chain requirements enforcement; SME engagement programme	Dynamic Runtime SBOMs and Digital Product Passports	Zero-Knowledge Proofs (ZKPs) & Pan-European Threat Intel sharing

Progression Matrix: Cognitive SecOps (LLM-SecOps) & Automated Threat Intelligence

Priority 7: Pan-European NaaS Federation (“Schengen for Cloud-Edge/AI”)

1. **Baseline & Target Objective:** *Baseline:* European telco deployments are fragmented and nation-bound. The lack of a true Digital Single Market prevents European operators from scaling to compete with US and Chinese hyperscalers. *Target:* A unified, interoperable European Network-as-a-Service (NaaS) platform enabling seamless “roaming for compute” across borders, legally operating as a single jurisdiction for data workloads.
2. **Primary Drivers:** *Market:* The “Sovereignty Cliff”—lack of a true Digital Single Market prevents European operators from achieving scale parity with hyperscalers. *Regulatory:* Data Act interoperability requirements; IPCEI-CIS mandates for cross-border infrastructure investment; CAMARA API standardization enabling federated network services.
3. **High Impact Activities:** Deployment of harmonized CAMARA APIs; automated compliance negotiation in 5G corridors; “Data Visa” mechanisms enabling dynamic cross-border AI

model migration; inter-operator East-West security interfaces; development of the “Schengen for Cloud” legal framework.

4. **Catalysts & Enablers:** IPCEI-CIS initial deployments; **European Alliance for Industrial Data, Edge and Cloud; CAMARA API alliance; Gaia-X** federation services; **ECCC** for cross-border security standards.
5. **Disruptors & Systemic Risks:** Conflicting national data sovereignty laws preventing true federation; regulatory arbitrage creating security-weakest-link vulnerabilities at cross-border interfaces; insufficient SLA enforcement mechanisms across operator boundaries.
6. **Failsafe Mechanisms & Resilience:** Graceful degradation protocols redirect workloads to national infrastructure if cross-border SLAs or compliance APIs fail; bilateral fallback agreements between operator pairs ensuring minimum viable connectivity.
7. **Transversal Meta-Layer:** *Sovereignty:* Enabling European-scale infrastructure that competes with US and Chinese hyperscalers without ceding control to non-EU providers. *Trust:* “Data Visas” providing legal clarity on data jurisdiction during cross-border migration. *Economic:* Creating a genuine Digital Single Mar-

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Siloed AI Copilots	Federated Edge Agents	Ubiquitous AI Immune System
SSL	High Reliance on US Foundation Models	Sovereign European Edge LLMs, SLMs, LCMs	Global Leadership in Trustworthy AI
Operational State	LLMs assist human SOC analysts; NWDAF telemetry routed to centralized AI for anomaly detection.	Edge-native “Guardian” nodes autonomously execute Zero-Trust reconfigurations and throttle threats.	Swarm-based agentic AI defends the Edge-Cloud Continuum via decentralized, proactive threat intelligence.
Critical Enabler	EU AI Act enforcement & EuroHPC AI Factories	Standardized AI Interconnect APIs	Self-Evolving AI Swarm Protocols

ket that enables European SMEs and operators to compete at continental scale.

Progression Matrix: Pan-European Naas Federation

Priority 8: Confidential Computing & Privacy-Preserving AI Pipelines

- Baseline & Target Objective:** *Baseline:* Basic encryption at-rest and in-transit is standard, but data-in-use remains exposed during processing. Multi-tenant edge nodes and federated AI training create significant insider threat and IP leakage risks. *Target:* Ubiquitous data-in-use protection via Trusted Execution Environments (TEEs) and Homomorphic Encryption across the full Cloud-to-IoT path, enabling collaborative AI without raw data ever leaving sovereign jurisdictions.
- Primary Drivers:** *Threat:* Insider threats in multi-tenant edge nodes; IP exfiltration during federated AI training across competitive organizational boundaries. *Regulatory:* **GDPR** stringency for data-in-use; **EUCS “High+”** certification requirements mandating confidential computing for sensitive cloud workloads; **AI Act** requirements for data governance in high-risk AI systems.
- High Impact Activities:** TEE deployment

on critical cloud nodes for EUCS compliance; sovereign hardware pilots using EU-developed RISC-V TEEs; multi-party privacy-preserving AI pipelines across multi-provider nodes; operationalization of Fully Homomorphic Encryption (FHE) for production analytics workloads.

- Catalysts & Enablers:** **EUCS** Certification Rollout; **European Data Spaces** as the deployment vehicle; **Chips JU** for RISC-V hardware development; **ECCC** strategic funding for FHE maturation.
- Disruptors & Systemic Risks:** TEE side-channel vulnerabilities rendering current enclave implementations obsolete; FHE performance overhead remaining prohibitive for real-time edge applications; continued dependency on non-EU TEE hardware (Intel SGX, ARM TrustZone).
- Failsafe Mechanisms & Resilience:** Limit processing to local, on-device federated learning nodes (never sharing raw data) if edge TEE integrity cannot be cryptographically verified; multi-TEE implementation diversity to avoid single-vendor hardware vulnerabilities.
- Transversal Meta-Layer:** *Sovereignty:* Moving from reliance on foreign hardware TEEs to sovereign RISC-V TEEs. *Trust:* Enabling European citizens and businesses to share data for AI without sacrificing privacy. *Ethics:* Providing

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Bilateral Interconnection	Federated Edge Roaming	True Digital Single Market
SSL	Hyperscaler Platform Dominance	Pan-European Consortium Competes	European Scale Parity Achieved
Operational State	Harmonized CAMARA APIs deployed; Automated Compliance Negotiators tested in 5G corridors.	“Data Visas” enable dynamic cross-border migration of AI models without legal friction.	Single unified computing continuum legally operates as one jurisdiction for data workloads.
Critical Enabler	IPCEI-CIS Initial Deployments	Inter-Operator East-West Interfaces	“Schengen for Cloud” Legal Framework

the technical foundation for Privacy-by-Design as required by GDPR.

Progression Matrix: Confidential Computing & Privacy-Preserving AI Pipelines

Priority 9: Resilient-by-Design Infrastructure & Autonomous “Island Mode”

1. **Baseline & Target Objective:** *Baseline:* Current cloud-dependent architectures have single points of failure. Edge nodes cannot function autonomously if WAN connectivity or centralized cloud services are disrupted. NIS2 mandates business continuity but infrastructure lacks native resilience capabilities. *Target:* Edge-native architectures capable of graceful degradation and autonomous local operation (“Island Mode”) during severe network, cloud, or power disruptions—ensuring continuity of critical services without any central cloud intervention.
2. **Primary Drivers:** *Threat:* Escalating global stressors—cyberattacks, extreme weather, geopolitical tensions—causing cascading failures across interconnected cloud-edge infrastructure. *Regulatory:* **NIS2** mandates for business continuity in critical infrastructures (healthcare, energy, transport); **DORA** (Digital Operational Resilience Act) requirements for financial sector; SNS JU requirements for 6G resilience and AI-RAN.
3. **High Impact Activities:** Edge node soft-state protocols enabling operation with cached credentials during cloud disconnections; AI-driven anomaly detection triggering graceful degradation automatically; dynamic spectrum sharing for emergency roaming; zero-energy backup layers maintaining minimal vital telemetry; bio-inspired elasticity for autonomous E2E recovery.
4. **Catalysts & Enablers:** **NIS2** & **DORA** Enforcement as market driver; **3GPP/ETSI** standardized fallback APIs; **SNS JU** 6G resilience research; **O-RAN** distributed open interfaces enabling autonomous RAN operation.
5. **Disruptors & Systemic Risks:** Security vulnerabilities introduced by offline-mode cached credentials being exploited; lack of standardized Island Mode protocols preventing interoperability during coordinated multi-site disruptions; power infrastructure failures disabling edge nodes before Island Mode activates.
6. **Failsafe Mechanisms & Resilience:** Zero-energy backup layers or localized Low-Power WAN (LPWAN) micro-slices maintaining a “minimal viable network” for life-saving telemetry if standard edge power and computing completely fail; local cryptographic credential caches with time-bounded validity to prevent indefinite offline operation.
7. **Transversal Meta-Layer:** *Sovereignty:* Ensuring European critical infrastructure can oper-

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Isolated Enclaves	Distributed Confidential Cloud	Ubiquitous Confidential Continuum
SSL	High Reliance on Foreign Hardware TPMs	European RISC-V TEEs Emerge	Total Data-in-Use Sovereignty
Operational State	TEEs deployed on critical cloud nodes for EUCS compliance; initial sovereign hardware pilots.	Multi-party privacy-preserving AI pipelines train across multi-provider nodes seamlessly.	Homomorphic encryption operationalized; end-to-end encrypted continuum from IoT to HPC.
Critical Enabler	EUCS Certification Rollout	RISC-V Hardware Scaling	Fully Homomorphic Encryption (FHE) Maturation

ate independently of any centralized or foreign-hosted cloud dependency. *Trust*: Guaranteeing continuity of essential services for European citizens during crises. *Sustainability*: Ultra-low-power Island Mode operation aligning resilience with Green IT objectives.

Progression Matrix: Resilient-by-Design Infrastructure & Autonomous “Island Mode”

Priority 10: Proactive Cyber Threat Management & Defense against Advanced Persistent Threats (APTs)

- Baseline & Target Objective:** *Baseline*: Intrusion Detection Systems are reactive with long “dwell times,” allowing APTs to persist undetected for months. Threat intelligence is siloed, national, and non-standardized. Incident response relies heavily on manual analyst workflows. *Target*: Proactive, AI-driven cyber threat hunting, multi-layered alert correlation, and dynamic Moving Target Defense (MTD) to disrupt the prolonged lifecycles of state-sponsored APTs before they achieve their objectives.
- Primary Drivers:** *Threat*: The increasing sophistication of nation-state actors blending cyber espionage with organized crime; exploitation of complex supply chains (e.g., SolarWinds-style attacks); use of stealthy techniques like

memory-only execution and legitimate administrative tools (“living off the land”) for lateral movement. *Regulatory*: NIS2 incident reporting timelines requiring faster detection and response; ECCC mandate for Pan-European Cyber Threat Intelligence (CTI) sharing.

- High Impact Activities:** SIEM/ML correlation of seemingly isolated events to reduce APT dwell times; machine-assisted threat hunting for Indicators of Compromise (IoCs) aligned with **MITRE ATT&CK**; LLMSecOps agents autonomously ingesting CTI and tracking anomalous lateral movement; AI-driven Moving Target Defense (MTD) continuously shuffling network resources and IP addresses to disrupt reconnaissance.
- Catalysts & Enablers:** **MITRE ATT&CK** framework integration into automated incident management; Machine Learning and **User and Entity Behavior Analytics (UEBA)** for continuous threat hunting; Pan-European CTI sharing platforms (CONCORDIA successor); **ECCC** coordination of national cyber intelligence capacities.
- Disruptors & Systemic Risks:** APT actors adapting faster than defensive AI can learn, using adversarial techniques to evade ML-based detection; legal barriers to real-time CTI sharing across Member State jurisdictions; AI-generated malware bypassing behavioral baselines trained

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Localized Fallback Piloting	Standardized Island Mode	Fully Autonomous Cyber-Physical Resilience
SSL	Vulnerable to Cascading WAN Failures	Contained Local Survivability	Systemic European Resilience
Operational State	Edge nodes use cached credentials and soft-state protocols to survive temporary cloud disconnections.	AI-driven anomaly detection triggers graceful degradation; dynamic spectrum sharing used for emergency roaming.	Bio-inspired elasticity; swarms seamlessly coordinate E2E cyber-physical recovery without any central cloud intervention.
Critical Enabler	NIS2 & DORA Enforcement	3GPP/ETSI Standardized Fallback APIs	“Cognitive Twin” E2E Orchestration

on historical attack patterns.

6. **Failsafe Mechanisms & Resilience:** Autonomous, real-time micro-segmentation (Zero-Trust) to instantly quarantine compromised supply chain components or sever Command and Control (C&C) pathways as soon as behavioral anomalies are detected, even if the specific zero-day exploit is unknown.
7. **Transversal Meta-Layer:** *Sovereignty:* Building European sovereign threat intelligence capacity reducing dependency on non-EU threat feeds and analysis platforms. *Trust:* Protecting European public institutions, critical infrastructure, and democratic processes from state-sponsored cyber operations. *Resilience:* Ensuring that even the most sophisticated adversaries face a continually shifting, proactive defense rather than a static target.

Progression Matrix: Proactive Cyber Threat Management & Defense against APTs

9 Executive Summary & Strategic Recommendations

Executive Summary

This cybersecurity and cyber-resilience strategy defines a comprehensive 15-year trajectory to se-

cure the European Cloud-Edge-IoT Continuum against emerging threats while establishing digital sovereignty. The strategy moves beyond traditional perimeter defenses toward autonomous, federated, and certified Security-by-Design architectures.

By utilizing a **4-Filter Prioritization Framework** (Regulatory Urgency, Foundational Dependency, Sovereignty Cliff, and Market Feasibility), ten strategic priorities are identified across three time horizons: **Hardware Roots of Trust & EUCC ‘High’ Certification**, **Post-Quantum Cryptographic Agility for Edge/IoT**, **Security Interoperability via Simpl Middleware**, **Supply Chain Verification**, **Cognitive SecOps (LLMSecOps)**, **Pan-European NaaS Federation**, **Confidential Computing**, **Resilient-by-Design Infrastructure**, and **Proactive APT Defense**. The three critical-path priorities that must be addressed immediately in Phase 1 (2025–2030) remain Hardware Roots of Trust, PQC Agility, and Simpl-based Security Interoperability, as they provide the foundational anchors upon which all subsequent capabilities depend.

The roadmap employs a dual-metric progression framework: **Operational Deployment Phase (ODP)** tracking market integration, and **Strategic Sovereignty Level (SSL)** measuring European independence from foreign technology dependencies. This enables objective tracking of capability maturity across three distinct phases: *Foundation & Compliance* (0–5 years), *Federation & Standardization* (5–10

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
ODP	Multi-Layered Alert Correlation & Hunting	Automated Triage & LLM-SecOps Containment	Ubiquitous Moving Target Defense (MTD)
SSL	High Dependency on Global Threat Feeds	Sovereign Pan-European CTI & Behavioral Analytics	Full Autonomous Counter-Adversarial Resilience
Operational State	SIEM systems and ML correlate isolated events to reduce APT dwell times; analysts conduct machine-assisted threat hunting for IoCs.	LLMSecOps agents autonomously ingest CTI, track anomalous lateral movement, and orchestrate rapid incident triage aligned with MITRE ATT&CK.	AI-driven MTD continuously shuffles network resources, IP addresses, and software diversity, actively disrupting APT reconnaissance and paralyzing attacks.
Critical Enabler	Federated SIEMs & Standardized CTI Sharing Protocols	LLMSecOps & UEBA (User and Entity Behavior Analytics)	Adaptive Edge-Cloud AI Swarms & Zero-Touch Automation

years), and *Autonomy & Leadership* (10–15 years).

Strategic Recommendations for Phase 1 (2025–2030)

To operationalize this strategy, the following actions are recommended for immediate European R&I funding and policy alignment:

- **Accelerate Sovereign Secure Silicon Development:** Immediate Horizon Europe and EU Chips Act funding for open-source RISC-V secure elements and hardware roots of trust engineered to meet **EUCC ‘High’ assurance levels**. Establish pilot fabrication lines dedicated to security-critical components to reduce 100% foreign dependency on TPMs and secure enclaves.
- **Operationalize Crypto-Agility for the Edge/IoT:** Develop and standardize automated crypto-agility middleware frameworks that allow resource-constrained IoT devices to upgrade to **Post-Quantum Cryptography (PQC)** without requiring hardware replacement. Prioritize ENISA guidelines alignment and ECCC funding for lightweight PQC implementations targeting 8/16-bit microcontrollers.
- **Enforce Security-by-Design via Simpl Integration:** Mandate standardized security **Minimal Interoperability Mechanisms**

(MIMs) across all Common European Data Spaces to ensure security policies are provider-agnostic, cryptographically verifiable, and eliminate vendor lock-in. Align Digital Europe Programme funding with Simpl stack security module development.

- **Harmonize Continuous Certification Life-cycles:** Transition from static, point-in-time PDF certificates to **Continuous Automated Assurance** frameworks using telemetry-driven compliance verification. Develop real-time certification tracking standards for both cloud services (**EUCS**) and physical edge products (**EUCC**), ensuring autonomous compliance maintenance throughout operational lifecycles.
- **Establish Federated Cyber Resilience Capabilities:** Invest in cross-border SOC federation infrastructure and legally protected threat intelligence sharing platforms to overcome institutional trust barriers between Member States. Align with CONCORDIA project outcomes and ECCC Strategic Agenda priorities for collaborative defense.
- **Synchronize Regulatory Enforcement Timelines:** Ensure coordinated enforcement of NIS2, Cyber Resilience Act (CRA), and AI Act requirements by 2026–2027 to create

unified market incentives for security investment. Provide targeted support for SMEs to meet compliance requirements without creating insurmountable barriers to market entry.

Success Metrics for 2030

By the end of Phase 1, Europe should achieve:

- **Hardware Sovereignty:** At least 30% of critical infrastructure edge nodes utilizing EU-designed, EUCC ‘High’ certified secure elements; first sovereign RISC-V secure element in industrial production.
- **Quantum Readiness:** 100% of new critical IoT deployments supporting hybrid (classical + PQC) schemes; complete cryptographic inventory of 80%+ legacy OT/IoT systems.
- **Federated Trust:** Majority of Common European Data Spaces operating on Simpl-based security interoperability with standardized MIMs adopted as CEN/CENELEC standards.
- **Certification Maturity:** Operational continuous certification frameworks for at least two major Data Spaces with real-time EUCC/EUCC tracking; point-in-time audit model deprecated in pilot sectors.
- **Cognitive SecOps Readiness:** LLM-assisted SOC tools deployed in at least five Member State national cybersecurity centres; NWDAF telemetry pipelines feeding centralized AI analytics for anomaly detection.
- **Supply Chain Transparency:** Runtime SBOM frameworks mandated for all new EU-funded cloud/edge deployments; Digital Product Passport pilots operational in the telecom and industrial IoT sectors.

Strategic Recommendations for Phase 2 (2030–2035)

Phase 2 transitions from piloting and compliance to *federation, standardization, and scaled sovereignty*.

The foundational hardware, cryptographic, and interoperability capabilities established in Phase 1 are industrialized and federated across Member States and sectors. The following recommendations constitute the critical-path actions required during this period.

- **Complete the PQC Migration for Critical Infrastructure:** Mandate hybrid PQC deployment for all public-sector ICT procurement by 2031 and deprecate classical-only cryptography for high-security contexts by 2033. Fund ECCC-led programmes to operationalize crypto-agility middleware across Industrial IoT, energy grids, and financial systems, building on the ENISA guidelines and NIST/ETSI finalization achieved in Phase 1. Initiate deployment of PQC-native RISC-V silicon for all new EU-manufactured edge infrastructure.
- **Scale Sovereign Secure Silicon to Industrial Dominance:** Leverage EU Chips Act fabrication milestones to transition RISC-V secure elements from pilot production to market-scale deployment, targeting 60%+ EU market share for secure silicon in critical edge infrastructure by 2035. Drive **EUCC ‘High’** certification to become the default procurement requirement across all Member State critical infrastructure procurements, eliminating foreign TPM dependency in regulated sectors.
- **Federate Security Across the Pan-European NaaS Ecosystem:** Deploy standardized CAMARA APIs and inter-operator east-west security interfaces to interconnect fragmented national telco edge clouds into a unified, security-audited Network-as-a-Service (NaaS) platform. Operationalize “Data Visa” frameworks enabling dynamic cross-border AI model migration without legal friction, and launch federated zero-trust policy enforcement spanning multiple national jurisdictions aligned with the IPCEI-CIS programme outcomes.
- **Deploy AI-Native, Autonomous SecOps at the Edge:** Transition SOC operations from

LLM-assisted copilot models (Phase 1) to **autonomous Edge-native Guardian nodes** capable of executing zero-trust reconfigurations and threat throttling without central cloud round-trips. Standardize AI Interconnect APIs under the EU AI Act high-risk system framework. Fund EuroHPC AI Factories to develop sovereign European edge LLMs and Small Language Models (SLMs) for security analytics, reducing reliance on US foundation models.

- **Operationalize Confidential Computing Across the Federated Continuum:** Scale TEE deployments from isolated cloud nodes (Phase 1) to **distributed confidential cloud** architectures spanning multi-provider nodes across Member States. Launch multi-party privacy-preserving AI pipelines for regulated sectors (health, finance, defence) that jointly compute over sensitive datasets without raw data leaving sovereign jurisdictions. Drive EUCS ‘High+’ certification requirements incorporating confidential computing as a baseline condition.
- **Establish a Quantum-Safe Data Space and QKD Backbone for Critical Links:** Operationalize the first fully end-to-end PQC-secured European Data Space by 2034. Deploy Quantum Key Distribution (QKD) for point-to-point optical links connecting government and critical infrastructure backbone nodes, aligned with the EU Quantum Flagship programme and 3GPP QKD specifications, providing information-theoretically secure key exchange complementary to PQC.
- **Industrialize Supply Chain Verification and Digital Product Passports:** Mandate Runtime SBOMs and Digital Product Passports (DPP) for all hardware and software components entering EU critical infrastructure supply chains. Deploy Zero-Knowledge Proof (ZKP) mechanisms for IP-protected cryptographic supply chain verification, enabling SMEs to demonstrate compliance without exposing proprietary implementation details. Transition from Phase 1’s disclosure-based model to **par-**

tial restriction of unverified supply chain components.

- **Activate Proactive APT Defense and Moving Target Defense (MTD):** Transition from SIEM-based reactive detection to **LLMSecOps-orchestrated autonomous triage:** AI agents that ingest Pan-European CTI feeds, track anomalous lateral movement aligned with the MITRE ATT&CK framework, and orchestrate rapid incident containment. Deploy User and Entity Behavior Analytics (UEBA) at scale across federated SOCs, and pilot Moving Target Defense (MTD) mechanisms in at least three Member State critical infrastructure sectors to actively disrupt APT reconnaissance.

Success Metrics for 2035

By the end of Phase 2, Europe should achieve:

- **Quantum-Resilient Infrastructure:** 100% of EU public-sector ICT using hybrid or full PQC schemes; first “Quantum-Safe Data Space” fully operational; classical-only cryptography deprecated in all high-security EU government systems.
- **Sovereign Silicon Dominance:** 60%+ EU market share for secure silicon in critical edge infrastructure; EUCC ‘High’ certification mandatory for regulated sector ICT procurement across all 27 Member States.
- **Federated NaaS Security:** Pan-European NaaS security framework operational across at least 10 telecom operators; “Data Visa” cross-border AI workload migration demonstrated in production at corridor scale; 70%+ Common Data Spaces using automated Simpl-based policy federation.
- **Autonomous AI SecOps:** Edge-native Guardian nodes autonomously executing zero-trust reconfigurations deployed in at least five Member State critical infrastructure sectors;

sovereign EU edge LLMs/SLMs operationally replacing non-EU foundation models in security analytics.

- **Confidential Computing at Scale:** Multi-party privacy-preserving AI pipelines operational in at least three cross-border regulated sectors; RISC-V TEEs scaling toward European silicon self-sufficiency for enclave hardware.
- **APT Dwell Time Reduction:** Mean APT dwell time reduced by 70% compared to Phase 1 baseline through AI-driven behavioral hunting; Pan-European CTI sharing platform operational with legally harmonized data exchange agreements covering all 27 Member States.

Strategic Recommendations for Phase 3 (2035–2040)

Phase 3 achieves *full sovereign autonomy and global leadership*. By this stage, European security architecture is self-sustaining, self-evolving, and serves as the global reference implementation for secure, trustworthy, and sustainable computing continuum infrastructure. The following recommendations define the critical-path actions to consolidate and export European leadership.

- **Achieve a Fully Quantum-Native European Computing Continuum:** Complete the migration to PQC-native silicon and cryptographic infrastructure across *all* EU Cloud-Edge-IoT tiers by 2036. Retire all hybrid classical/PQC transitional schemes in sovereign infrastructure and mandate PQC-by-default for all new EU ICT products. Position ETSI/CEN/CENELEC-led PQC standards as the global reference baseline, driving export of European quantum-resilience frameworks to allied nations and international standards bodies (ISO, ITU-T).
- **Establish Full Sovereign Autonomy in Secure Silicon Fabrication:** EU Chips Act fabrication capacity should meet 80%+ of European secure silicon demand by 2039, elimi-

nating residual critical dependencies on extra-territorial semiconductor supply chains. Open-source RISC-V secure enclave architectures certified at EUCC ‘High’ should become the default hardware foundation for all new EU ICT products across HPC, Cloud, Edge, and Device tiers, from consumer IoT to governmental systems.

- **Realise the “Schengen for Cloud-Edge” Legal and Technical Framework:** Operationalize the unified European computing continuum as a single jurisdiction for AI workload data governance, with legally binding cross-border SLA frameworks and harmonized compliance APIs (building on Phase 2 Data Visa pilots). The **EU Cloud Rulebook** should reach full legal maturity by 2037–2038, making provider-agnostic, zero-trust cross-border federation the normative baseline for all EU-funded digital infrastructure.
- **Deploy Swarm-Based Agentic AI Immune Systems:** Transition from Phase 2’s autonomous Edge Guardian nodes to fully **decentralized, self-evolving AI swarms** capable of proactively defending the Cloud-Edge Continuum through distributed threat intelligence, zero-touch automated incident response, and bio-inspired elasticity for E2E cyber-physical recovery. Governance frameworks for **AI Trustworthiness and Algorithmic Accountability** (CL4.7) must be operationalized to ensure that autonomous security agents themselves meet EU AI Act high-risk system requirements, with explainability guarantees for all autonomous containment decisions.
- **Achieve Ubiquitous Moving Target Defense and Autonomous Counter-APT:** AI-driven MTD should continuously and autonomously shuffle network resources, IP addresses, and software diversity across the entire EU continuum, actively paralyzing APT reconnaissance and lateral movement without human intervention. The Pan-European CTI platform should evolve into a **Full Autonomous Counter-Adversarial Resilience**

posture, with AI swarms proactively disrupting offensive operations rather than merely detecting and responding to them.

- **Operationalize Homomorphic Encryption and SMPC for Production Workloads:** Full Homomorphic Encryption (FHE) performance should reach operational viability for real-time analytics by 2037–2038, completing the trajectory from Phase 1 isolated enclaves through Phase 2 distributed confidential clouds to a **ubiquitous confidential continuum** from IoT sensor to HPC node. Secure Multi-Party Computation (SMPC) should be standardized for cross-border federated AI in regulated sectors (health, finance, defence, justice), enabling collaborative data science without any raw sensitive data ever crossing a sovereign boundary.
- **Complete Full Restriction of Non-Verified Supply Chain Components:** The supply chain verification framework should mature from Phase 2’s partial restriction model to **full automated blocking** of any unverified hardware, OS, or software component attempting to enter EU critical infrastructure. Zero-Knowledge Proof-based verification should be the standard cryptographic mechanism for IP-protected supply chain attestation. Pan-European threat intelligence on supply chain compromises should flow in real time across all Member State national agencies via the federated CTI platform.
- **Position Europe as the Global Standard-Setter for Trustworthy Security:** European cybersecurity standards—EUCS, EUCC, security MIMs, PQC migration frameworks, and continuous certification lifecycle models—should be actively promoted through bilateral agreements, multilateral standards bodies, and the EU’s external digital policy instruments (e.g., EU–US Trade and Technology Council, EU–Global Gateway). The goal is for European frameworks to become the global reference implementation for *trustworthy, sovereign, and sustainable* computing security, generating export value for the European cybersecurity industry and reducing

global dependency on non-democratic technology governance models.

Success Metrics for 2040

By the end of Phase 3, Europe should achieve:

- **Full Quantum-Native Continuum:** 100% of EU Cloud-Edge-IoT critical infrastructure natively running PQC; EU PQC and QKD standards adopted as ISO/ITU-T global baseline; no operational classical-only cryptography in sovereign EU systems.
- **Sovereign Silicon Self-Sufficiency:** 80%+ of European secure silicon demand met by EU Chips Act fabrication capacity; RISC-V EUCC ‘High’ the default hardware foundation for all new EU ICT products.
- **Unified European Digital Jurisdiction:** EU Cloud Rulebook fully enforced; 100% of Common Data Spaces operating on provider-agnostic zero-trust architectures; “Schengen for Cloud-Edge” legally operational across all 27 Member States.
- **Autonomous AI Security Fabric:** Self-evolving AI swarm defense deployed across the EU continuum; all autonomous security agents compliant with EU AI Act Trustworthiness and Accountability requirements; Security-by-Design the normative default—manual security configuration obsolete.
- **Ubiquitous Confidential Computing:** FHE operationalized for real-time production analytics; SMPC standardized for cross-border federated AI in all regulated sectors; end-to-end encrypted continuum from battery-less IoT to HPC without any raw sensitive data leaving sovereign boundaries.
- **Full Counter-APT Resilience:** AI-driven MTD continuously disrupting APT reconnaissance in real time across the full continuum; Pan-European CTI platform providing proactive counter-adversarial intelligence; mean APT

dwel time reduced to near-zero for known attack pattern families.

- **Global Leadership:** European cybersecurity frameworks serving as the global reference implementation; EU cybersecurity technology export value exceeding pre-roadmap baseline by a measurable factor, as assessed by the ECCC and ENISA in the 2040 roadmap completion review.

10 Discussion

The proposed methodology bridges a gap between architecture-centric security analysis and policy-oriented R&I prioritization. Its main strength lies in explicit traceability from candidate capability to executive recommendation, using structured filters, hybrid profiling dimensions, and phase-based progression logic.

From an IEEE Access perspective, the framework can be interpreted not only as a strategic roadmap artifact but also as a reusable methodological contribution for evaluating sovereign cybersecurity capabilities in heterogeneous computing continua.

11 Conclusion

This paper presented a structured and traceable methodology for cybersecurity prioritization in the Cloud–Edge–IoT continuum. By integrating architectural modeling, multi-dimensional prioritization, and phased progression analysis, the framework enables evidence-based decision-making aligned with European digital sovereignty objectives.

Future work includes quantitative validation, integration with real deployment data, and extension toward automated policy-driven orchestration.

References

- [1] European Parliament and Council, “Directive (EU) 2022/2555 on measures for a high common level of cybersecurity across the Union (NIS2 Directive),” *Official Journal of the European Union*, Dec. 2022.
- [2] European Commission, “Proposal for a Regulation on horizontal cybersecurity requirements for products with digital elements and amending Regulation (EU) 2019/1020 (Cyber Resilience Act),” COM(2022) 454 final, Sep. 2022.
- [3] European Parliament and Council, “Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act),” *Official Journal of the European Union*, Jun. 2024.
- [4] European Union Agency for Cybersecurity (ENISA), “European Cybersecurity Certification Scheme for Cloud Services (EUCS),” Candidate Scheme, Dec. 2020.
- [5] European Union Agency for Cybersecurity (ENISA), “EUCC, a candidate cybersecurity certification scheme to serve as a successor to the existing SOG-IS,” V1.0, Jul. 2020.
- [6] European Alliance for Industrial Data, Edge and Cloud, “European industrial technology roadmap for the next-generation cloud-edge,” European Commission, 2024.
- [7] Smart Networks and Services Joint Undertaking (SNS JU), “Strategic Research and Innovation Agenda (SRIA),” 2023.
- [8] European Cybersecurity Competence Centre (ECCC), “ECCC Strategic Agenda for cybersecurity industrial, technology and research competence,” 2024.
- [9] G. Dreo, C. Schmitt, A. van der Wees, and N. Suri, “Deliverable D4.4: Cybersecurity Roadmap for Europe,” *CONCORDIA Cyber Security Competence for Research and Innovation (Horizon 2020)*, Dec. 2022.